

Project Proposal

**Open University Short Course Website for OUSL
Kandy Regional Centre (KRC)**

By
Group 71

SUPERVISOR.

Dr. (Ms.) W.A.L. Niwanthi,

Senior lecturer

Mathematics and Philosophy of Engineering,
Faculty of Engineering Technology,
The Open University of Sri Lanka.

Course Coordinator

Dr. (Mrs) K.V.J.P. Ekanayake

Senior lecturer

Dept. of Electrical and Computer Engineering

Bachelor of Software Engineering (BSE) Honours.

Dept. of Electrical and Computer Engineering

The Open University of Sri Lanka.

2026.

This Project Proposal is submitted to the Dept. of Electrical and Computer Engineering, The Open University of Sri Lanka, to fulfil the Mini project Component of Second Year Bachelor of Software Engineering (BSE) Honours.

DECLARATION

We, undersigned, and declare that the mini project on **Open University Short course web site (KRC)**, is our own work and that all sources we have used have been indicated and acknowledged by mean of complete references, and that this work has not been submitted before for any other degree at any other institution.

We are writing to kindly request your approval for our project titled “Open University Short Course Website for OUSL – Kandy Regional Centre (KRC)”.

The project aims to develop a centralized web-based platform through which students and visitors can easily access short-course details, schedules, and registration information, while improving the process of accessing and managing such information.

We are Group No. 71 for the 4189 subject, and we kindly request your approval to proceed with this project under your supervision.

Thank you for your time and consideration. We look forward to your valuable feedback and approval.

Yours sincerely,

Group Number : 71.

Date of Submission : 22/08/2026.

This Mini project Proposl has been submitted for evaluvation with supervisors’ approval.

SUPERVISORS’ APPROVAL

Dr.(Ms.) W.A.L. Niwanthi)

Senior lecturer

Mathematics and Philosophy of Engineering,

Faculty of Engineering Technology,

The Open University of Sri Lanka.

APPROVE .
Signature:
27/08/2026

Contents

1.1. BACKGROUND.....	7
2. PROBLEM STATEMENT & PROJECT OBJECTIVES	8
3. SIMILAR EXISTING PROBLEMS	9
4. PROPOSED SOLUTION	10
4.1. FLOW CHART	11
5. TECHNOLOGY PLAN (BACKEND , FRONTEND & AI TOOLS)	12
6. PROJECT TIMELINE AND CONCLUSION	13
7. PRELIMINARY SOFTWARE REQUIREMENTS SPECIFICATION (SRS)	15
7.1. INTRODUCTION.....	15
7.1.1. Purpose	15
7.1.2. Summary	15
7.1.3. Company Overview.....	15
7.1.4. Team Members.....	16
7.1.5. Project Overview	16
7.1.6. Scope	17
7.1.7. Assumptions	17
7.1.8. Definitions, Acronyms and Terminology.....	17
7.1.9. References	18
7.2. PROJECT SCOPE AND IMPACT	18
7.2.1. Context Diagram	18
7.2.2. Scope of the Project.....	19
7.2.3. Scope Exclusions.....	19
7.2.4. Impact on Other Systems	20
7.3. FUNCTIONAL REQUIREMENTS	20
7.3.1. User Access and Engagement	21
7.3.2. Information and Context Management	21
7.3.3. Education Resources and Tools	21
7.3.4. User Interaction and Support.....	22
7.3.5. Contribution to the University.....	22
7.3.6. Data Archival And Retention.....	22
7.4. USER PROFILES ,ROLES AND PRIVILEGES.....	23
7.4.1. Use Case Diagram.....	27

7.4.2.	Main Actors of the System.....	28
7.4.3.	7.4.4. Reporting Requirements.....	28
7.5.	NON FUNCTIONAL REQUIREMENTS	29
7.6.	USER INTERFACES	31
7.6.1.	WireFrame.....	32
7.7.	HARDWARE INTERFACES.....	38
7.7.1.	Server Requirements	38
7.7.2.	Client Devices	38
7.7.3.	Web Application Server	38
7.7.4.	Database Server	39
7.7.5.	Network Interface.....	39
7.8.	SOFTWARE INTERFACES	39
7.8.1.	Operating System Requirements	39
7.8.2.	Development Tools and Software	39
7.9.	COMMUNICATION INTERFACES.....	40
7.10.	SECURITY AND AUTHENTICATION REQUIREMENTS.....	40
7.10.1.	User Authentication.....	41
7.10.2.	User Registration.....	41
7.10.3.	Password Security	41
7.10.4.	Password Reset.....	41
7.10.5.	Role-Based Access Control.....	41
7.10.6.	Authorization.....	41
7.10.7.	Secure Communication	41
7.10.8.	Session Management.....	42
7.10.9.	Input Validation.....	42
7.10.10.	Account Protection.....	42
7.10.11.	Data Privacy	42
7.10.12.	Administrative Security.....	42
7.10.13.	Backup and Recovery.....	42
7.10.14.	Security Monitoring	42
8.	APPENDIX	43
8.1.	LETTER REQUESTING APPROVAL	43

Figure 1 Proposed Solution	11
Figure 2: Project Time Line	14
Figure 3: Context Diagram.....	18
Figure 4: Use Case Diagram	27
Figure 5: Homepage wireframe (Guest User) -course browsing, search, and login/register	32
Figure 6: Administration dashboard wireframe -user management, courses, enrollment tracking, and reports	32
Figure 7: Administration dashboard wireframe -user management, courses, enrollment tracking, and reports	33
Figure 8: Login / register screen (Guest User).....	33
Figure 9: Resources screen — downloadable materials and recordings per enrolled course	34
Figure 10: Profile screen - editable personal details and password change	34
Figure 11: Manage users screen -account list with role and status controls	35
Figure 12: Courses and programmes screen — course list with enrollment counts	35
Figure 13: Content and announcements screen — published items with edit controls.....	36
Figure 14: Reports and analytics screen -enrollment trend and downloadable reports.....	36
Figure 15: System settings screen - site-wide configuration fields.....	37
Figure 16: Letter Requesting Approva.....	43
 Table 1: Team Members	 16
Table 2: User Profiles ,Roles and Privileges.....	26
Table 3: Non Functional Requirements.....	30

INTRODUCTION

1.1. Background

The Open University Short Course Website for the Open University of Sri Lanka (OUSL) – Kandy Regional Centre (KRC) is a web-based information and registration management system developed to provide an efficient, reliable, and user-friendly platform for managing short-term educational programs. The website is designed to support individuals who wish to improve their knowledge and professional skills by enrolling in various short courses offered by the Kandy Regional Centre. The primary objective of the system is to simplify and modernize the traditional manual process of course promotion and registration. Instead of requiring applicants to visit the regional centre to obtain course details, collect application forms, or submit documents, the website enables them to complete these tasks conveniently from any location with internet access. This digital approach saves time, reduces paperwork, minimizes administrative effort, and enhances the overall user experience.

The website provides a comprehensive catalogue of short courses offered by the Kandy Regional Centre. Users can browse courses based on different categories such as Information Technology, Business Management, Languages, Education, Engineering, Health Sciences, and other professional development programs. Each course page contains detailed information, including the course title, description, learning objectives, expected learning outcomes, course duration, lecture schedules, mode of delivery (physical, online, or blended), registration opening and closing dates, commencement dates, course fees, payment methods, eligibility requirements, available seats, assessment methods, and certificate information. This detailed information helps prospective learners compare different courses and make informed decisions before applying.

Overall, the Open University Short Course Website significantly enhances the accessibility, visibility, and management of short-term educational opportunities offered by the Open University of Sri Lanka – Kandy Regional Centre. By providing a secure, modern, and user-friendly online platform for course information, registration, and communication, the system improves operational efficiency, reduces administrative workload, minimizes manual errors, increases transparency, and delivers a more convenient and satisfying experience for both prospective learners and university staff. It supports the university's mission of expanding access to quality education through technology while promoting lifelong learning and professional development within the community.

2. Problem Statement & project Objectives

The Open University of Sri Lanka (OUSL) – Kandy Regional Centre (KRC) provides various short-term educational and professional development programs for individuals seeking to improve their knowledge and skills. However, the current process of promoting and managing these short courses still relies on traditional methods such as printed advertisements, emails, telephone inquiries, and physical visits to the regional centre. Due to the absence of a dedicated online platform, prospective learners often experience difficulties in accessing complete and updated information about available courses, including course descriptions, learning outcomes, schedules, fees, eligibility requirements, registration deadlines, and application procedures. This may cause delays in decision-making and result in learners missing suitable course opportunities.

The existing manual registration process also creates challenges for both applicants and university staff. Applicants need to visit the regional centre to collect forms, submit documents, and obtain course-related information, which can be inconvenient and time-consuming. For administrative staff, managing paper-based applications and maintaining manual records increases workload and the risk of errors such as incorrect data entry, duplicate records, and difficulties in tracking applications. Additionally, the lack of a centralized system makes it challenging to update course information, monitor registrations, share announcements, and generate reports efficiently.

To address these issues, this project aims to develop a web-based Open University Short Course Website for the OUSL – Kandy Regional Centre. The proposed system will provide a centralized digital platform where learners can easily explore available courses, view detailed course information, create user accounts, complete online registrations, upload required documents, and track their application status. The system will also support administrators by allowing them to manage course details, update schedules, publish announcements, monitor registrations, and maintain learner records through a secure management interface.

The proposed website will improve accessibility, communication, and operational efficiency by reducing paperwork, minimizing manual errors, and simplifying the registration process. With a responsive and user-friendly design, learners can access the platform through different devices at any time, while university staff can manage course-related activities more effectively. Overall, the system will enhance the visibility and management of short courses offered by OUSL – Kandy Regional Centre and support the university's goal of expanding access to quality education through digital transformation.

3. Similar Existing Problems

Several educational institutions and online learning platforms currently use web-based systems to manage short courses, training programs, and student registrations. These existing systems provide useful features such as online course information, user registration, payment management, and communication facilities, which can be considered as references for developing the Open University Short Course Website for OUSL – Kandy Regional Centre.

One similar system is the Open University of Sri Lanka (OUSL) main website, which provides general information about academic programs, courses, admission procedures, and regional centres. However, it mainly focuses on overall university services and does not provide a dedicated platform specifically designed for managing short course details, online registrations, and short-term program administration.

Another example is the Learning Management Systems (LMS) used by universities and educational organizations. These platforms allow students to access course materials, communicate with instructors, submit assignments, and monitor learning progress. However, most LMS platforms focus on course delivery after enrollment rather than providing a complete solution for short course promotion, registration, and applicant management.

Many private educational institutes also use online course registration platforms where users can browse available programs, check course details, register online, and receive confirmation notifications. These systems improve accessibility and reduce manual work, but they may not include specific administrative features required by a university regional centre, such as managing multiple short courses, maintaining applicant records, and generating enrollment reports.

Compared with these existing systems, the proposed Open University Short Course Website aims to provide a more specialized solution for the Kandy Regional Centre by combining course information management, online registration, communication, and administrative functions within a single platform. The system will address the limitations of existing approaches by providing accurate course information, reducing manual processes, improving communication between learners and staff, and creating a more efficient and user-friendly experience for managing short-term educational programs.

4. Proposed Solution

The proposed solution is to develop a web-based Open University Short Course Website for the Open University of Sri Lanka (OUSL) – Kandy Regional Centre (KRC) to improve the management, promotion, and registration process of short-term educational programs. The system will provide a centralized online platform that allows prospective learners to easily access course information, register for courses, and communicate with the regional centre without depending on manual processes.

The proposed system will include two main user roles: Learners and Administrators. Learners will be able to create accounts, log into the system, browse available short courses, search courses by category or keywords, view detailed course information, check eligibility requirements, submit online applications, upload necessary documents, receive registration confirmations, and track their application status. This will provide a convenient and efficient way for learners to discover and enroll in suitable short courses.

Administrators will have access to a secure management dashboard where they can manage course information and student registrations. They will be able to add new courses, update or remove existing course details, manage registration periods, upload schedules, publish announcements, review applications, maintain learner records, and generate reports related to course registrations. This will reduce administrative workload, improve data accuracy, and support better decision-making. Overall, the proposed solution transforms the existing manual short course management process into a modern digital system. It improves accessibility, reduces paperwork, minimizes errors, enhances communication, and provides a more transparent and user-friendly experience for both learners and university staff.

4.1. Flow Chart

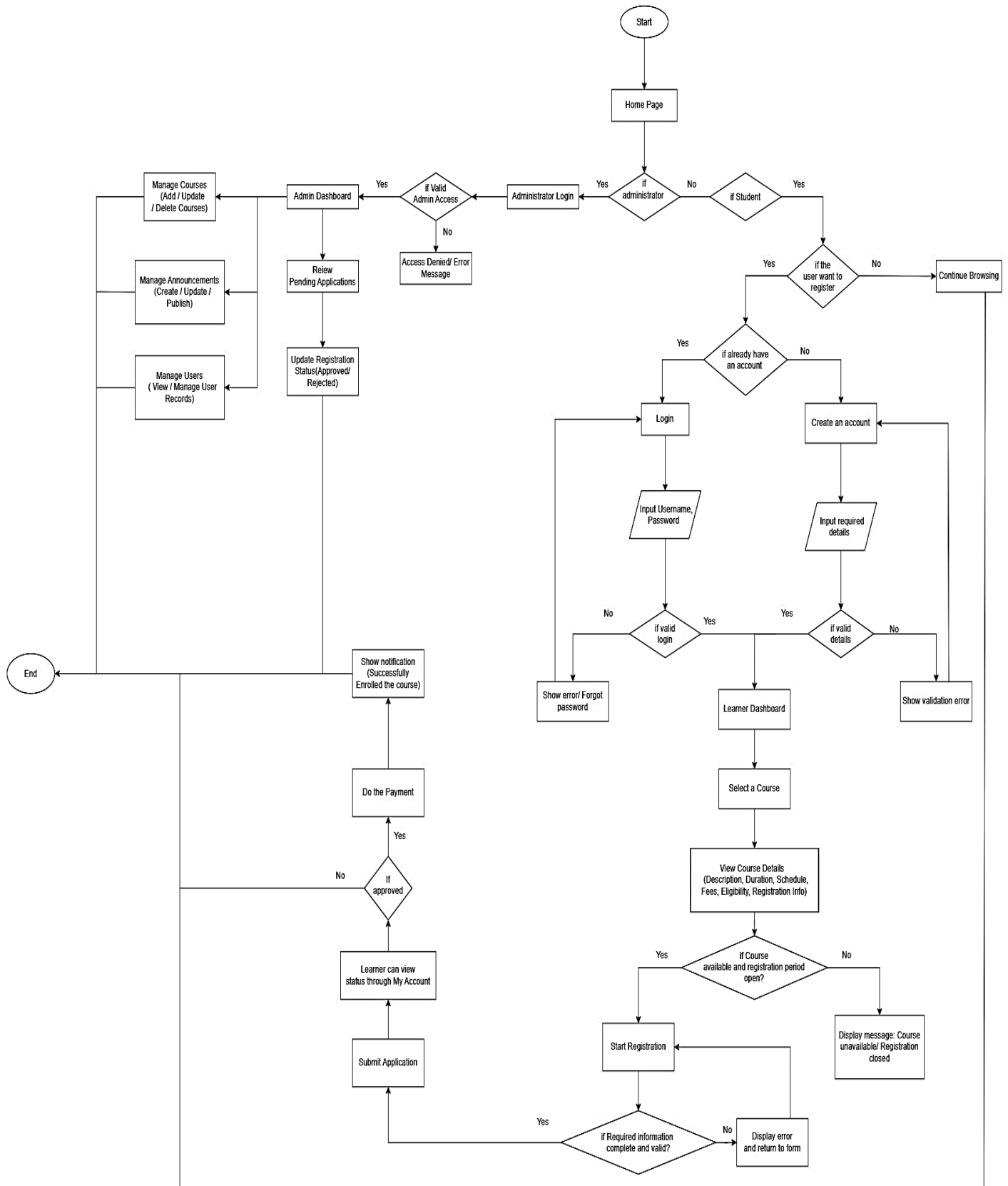


Figure 1 Proposed Solution

5. Technology Plan (Backend , Frontend & AI Tools)

The proposed **Open University Short Course Website** will be developed using the **MERN Stack (MongoDB, Express.js, React.js, and Node.js)** to provide a modern, scalable, and efficient web application. The backend of the system will be developed using **Node.js** with the **Express.js framework**, which will handle server-side operations, API development, user authentication, course management, registration processing, and communication between the frontend and database. **MongoDB** will be used as the database technology to store and manage user details, course information, registration records, documents, and administrative data in a flexible and efficient manner.

The frontend of the system will be developed using **React.js**, a JavaScript-based library that enables the creation of interactive, responsive, and user-friendly interfaces. React components will be used to design different sections of the website, including course browsing pages, registration forms, user dashboards, and administrative panels. Additional frontend technologies such as **HTML5, CSS3, JavaScript, Bootstrap, and responsive design techniques** will be used to improve usability and ensure compatibility across different devices.

For **UI/UX design**, **Figma** will be used to create wireframes, prototypes, and user interface designs before implementation. Figma will help visualize the website layout, user navigation flow, color schemes, and overall user experience. This design process ensures that the final system provides an intuitive, accessible, and visually appealing experience for both learners and administrators.

The development process will utilize tools such as **Visual Studio Code** as the primary code editor, **Git and GitHub** for version control and collaboration, **Postman** for API testing, and browser developer tools for debugging and performance optimization. These tools will support efficient development, testing, and maintenance of the system.

Generative AI tools will also be incorporated during the development process to improve productivity and assist with different project activities. AI tools may be used for generating coding suggestions, debugging assistance, documentation support, UI/UX improvement ideas, test case generation, and requirement analysis support. However, all AI-generated outputs will be reviewed and customized to ensure accuracy, security, and alignment with the project requirements.

Overall, the selected technology stack provides a complete solution for developing a secure, responsive, and user-friendly short course management platform while supporting efficient development, future scalability, and enhanced user experience.

6. Project Timeline and Conclusion

The development of the **Open University Short Course Website for the Open University of Sri Lanka – Kandy Regional Centre (KRC)** is planned over a **16-week period** following the Software Development Life Cycle (SDLC). During **Weeks 1–2**, the project focuses on requirement gathering, stakeholder analysis, problem identification, and preparation of the project proposal. **Weeks 3–4** are allocated for requirements elicitation, analysis, validation, and the preparation of the Software Requirements Specification (SRS). In **Weeks 5–6**, the system design phase includes UI/UX design using Figma, database design, and the creation of use case and system models. The **backend development** takes place during **Weeks 7–11** using **Node.js, Express.js, and MongoDB**, where APIs, user authentication, course management, and registration functionalities are implemented. During **Weeks 12–14**, the **frontend** is developed using **React.js**, focusing on responsive user interfaces, dashboards, and system integration. **Week 15** is dedicated to comprehensive system testing, bug fixing, performance optimization, and user acceptance testing, while **Week 16** covers final documentation, system deployment, project presentation, and submission. The project milestones include the approval of the project proposal, completion of the SRS, UI/UX prototype, backend and frontend development, successful system testing, and the final deployment and submission of the completed system.

In conclusion, the Open University Short Course Website will digitize the manual course management process by providing an efficient platform for learners and administrators. Developed using the MERN Stack, the system will improve registration, course management, communication, and operational efficiency while supporting future enhancements such as payments, analytics, and AI-based services.

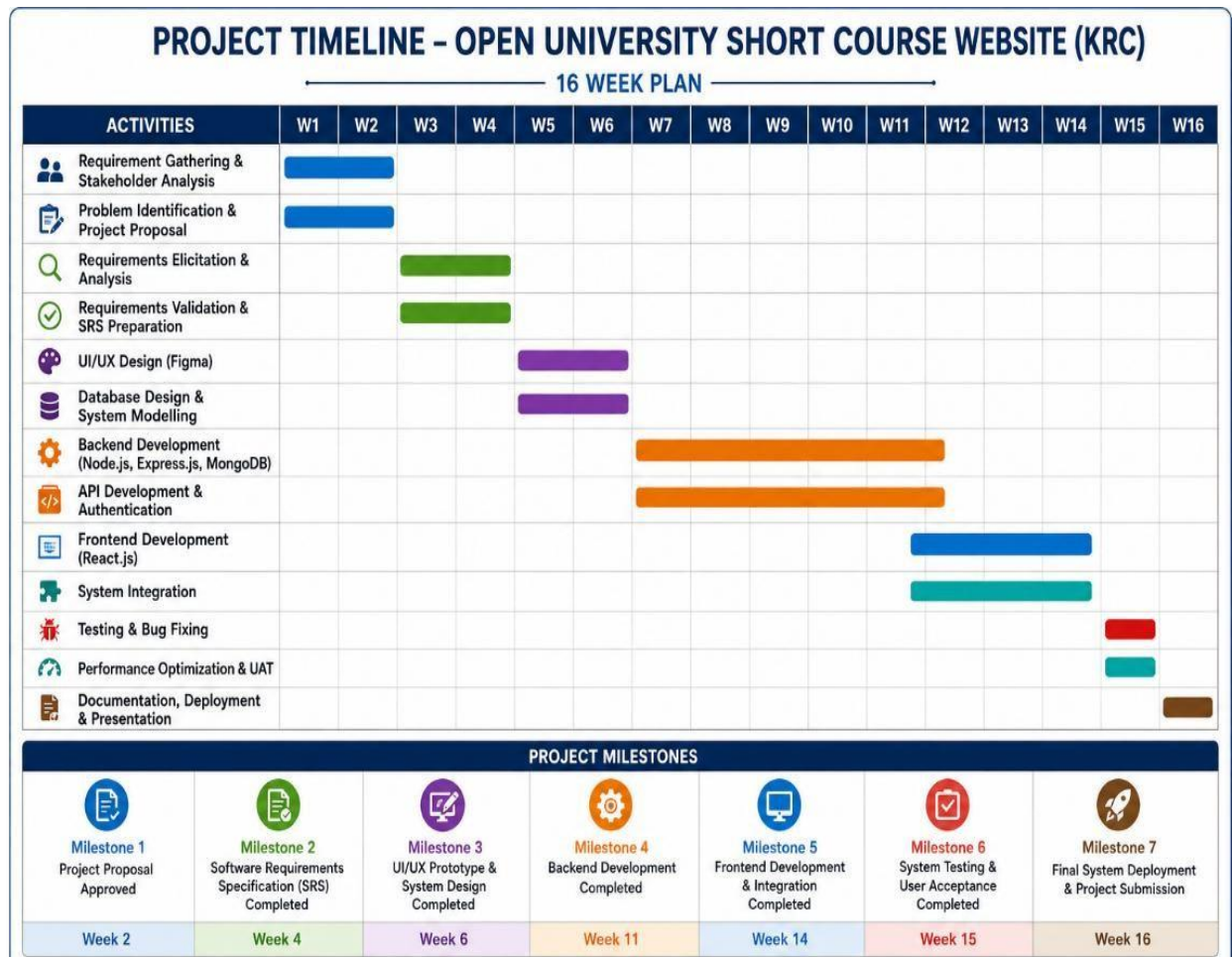


Figure 2: Project Time Line

7. Preliminary Software Requirements Specification (SRS)

7.1. Introduction

7.1.1. Purpose

The purpose of this document is to define the software requirements for the **Open University Short Course Website (KRC)**. The system provides a centralized platform for users to browse course details, schedules, fees, eligibility, and registration information. It aims to replace the traditional manual process with an efficient online platform while clearly identifying the requirements of learners and university staff.

7.1.2. Summary

The **Open University Short Course Website (KRC)** aims to modernize the traditional process of promoting, applying for, and registering for short courses at the **Open University of Sri Lanka – Kandy Regional Centre**. The website provides a centralized platform where learners can easily access course details, schedules, fees, eligibility requirements, and registration information. By improving access to accurate and up-to-date course information, the system will make the registration process more convenient, efficient, and accessible for learners while supporting university staff in managing short courses effectively.

7.1.3. Company Overview

Our project team is dedicated to developing a modern and user-friendly digital platform for the Open University of Sri Lanka – Kandy Regional Centre (KRC). The team focuses on using technology to simplify the promotion, application, and registration processes for short courses through the Open University Short Course Website. With a strong commitment to improving accessibility and efficiency, we aim to provide learners with accurate course information while supporting university staff in effective course management.

7.1.4. Team Members

Sr. No.	Name	SID	Registration No.	Contact No.
1.	S.M.I.I.K Wimalasekara	S92068547	321428547	0785653777
2.	M.M.A.G.S.A Warnathilake	S22010276	422514144	0705766401
3.	R.K.G.T.D Gnanasekara	S22010687	622516105	0702709680
4.	K Pushpakanthan	S92061424	121421424	0755415479

Table 1: Team Members

7.1.5. Project Overview

The **Open University Short Course Website (KRC)** is a web-based platform designed to provide learners with centralized access to information about short courses offered by the **Open University of Sri Lanka – Kandy Regional Centre (KRC)**. The system aims to modernize the traditional process of course promotion, information sharing, application, and registration by providing a convenient and efficient online solution.

The **home page** serves as the main gateway to the website, displaying available short courses, important announcements, upcoming courses, and other relevant information. Users can **register and log in** to access personalized features, while guests can browse available course information with limited access.

The **course page** provides detailed information about each course, including course descriptions, duration, schedules, fees, eligibility requirements, and available enrollment options. The **registration feature** allows learners to apply for selected courses online, reducing the need for manual applications. Users can also view their registration details through their accounts.

The website includes a **Forgot Password** feature that allows registered users to securely reset their passwords when necessary. The **My Account** section enables users to manage their personal information and account details. University staff or administrators can manage course information, registrations, announcements, and other system content through an administrative interface.

The **About Us** page provides information about the Kandy Regional Centre and its short-course services, while the **Contact Us** page allows learners to communicate with the university and obtain assistance.

Overall, the **Open University Short Course Website (KRC)** aims to improve accessibility, efficiency, and transparency in short-course management by bringing course information and registration services into one centralized online platform. This will provide a better experience for learners while reducing the administrative workload of university staff.

7.1.6. Scope

This document covers the **functional and non-functional requirements** for the **Open University Short Course Website (KRC)**. It details the required features, performance standards, compatibility requirements, security measures, usability, and quality assurance criteria necessary to develop a reliable, efficient, and user-friendly platform for managing and registering for short courses.

7.1.7. Assumptions

- 1) Users will have basic internet access and computer literacy.
- 2) The website will be accessible on **desktop, tablet, and mobile devices**.
- 3) Course information, schedules, fees, and announcements will be regularly updated by **authorized university administrators**.
- 4) Users will provide accurate information when registering for courses.
- 5) University staff will have the necessary access and permissions to manage course and registration details.

7.1.8. Definitions, Acronyms and Terminology

Term	Definition
OUSL	Open University of Sri Lanka
KRC	Kandy Regional Centre
SCS	Short Course System
UI	User Interface
UX	User Experience
SSL	Secure Sockets Layer
DB	Database
SRS	Software Requirements Specification
Admin	Authorized university staff responsible for managing the website
User	A learner or visitor who accesses the website

7.1.9. References

1. Open University of Sri Lanka (OUSL) – Official Website.
2. Open University of Sri Lanka – Kandy Regional Centre (KRC) information and short-course materials.
3. Software Requirements Specification (SRS) guidelines and documentation standards.
4. React.js Documentation – User interface development reference.
5. Node.js and Express.js Documentation – Backend development reference.
6. MongoDB Documentation – Database management reference.
7. Figma Documentation – UI/UX design reference.
8. OWASP Web Security Guidelines – Web application security reference.

7.2. Project Scope and Impact

The **Open University Short Course Website (KRC)** aims to modernize the management of short courses at the **Open University of Sri Lanka – Kandy Regional Centre** by providing a secure, reliable, and user-friendly online platform. By improving access to course information, online registration, and communication, the system will make the learning and application process more convenient for students. It will also help university staff manage courses and registrations more efficiently. Overall, the project aims to improve the quality and accessibility of short-course services while reducing the limitations of the traditional manual process.

7.2.1. Context Diagram

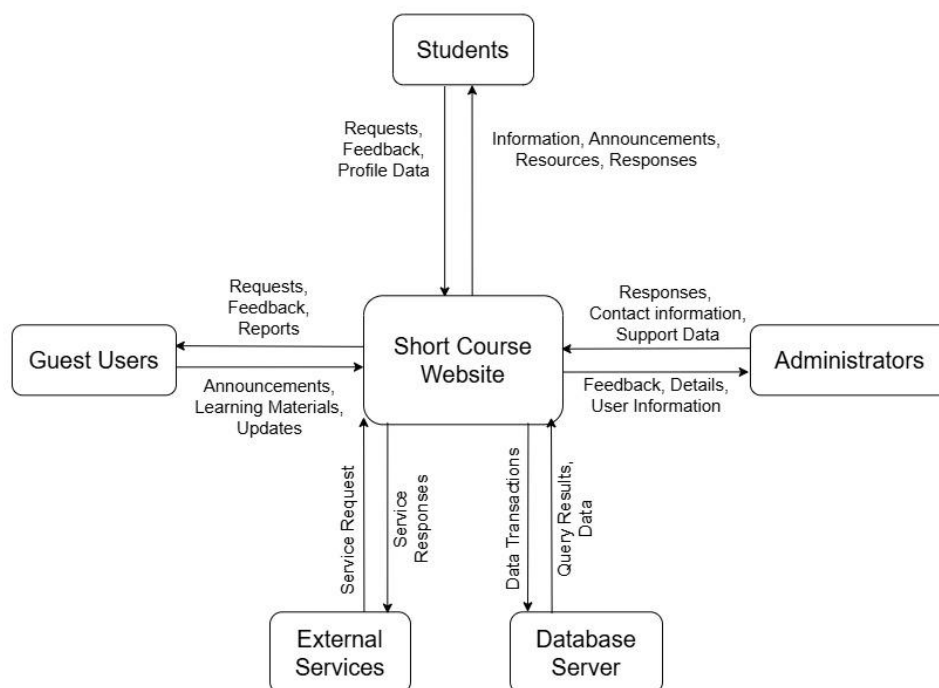


Figure 3: Context Diagram

7.2.2. Scope of the Project

The scope of the **Open University Short Course Website (KRC)** includes the development and deployment of a centralized web-based platform with the following features:

- 1) **Home Page:** Provides an overview of available short courses, latest announcements, upcoming courses, and important links.
- 2) **Course Information Page:** Provides detailed information about short courses, including course descriptions, duration, schedules, fees, and eligibility requirements.
- 3) **Course Registration:** Allows learners to apply and register for selected short courses through an online process.
- 4) **User Registration and Login:** Allows users to create accounts, securely log in, and access personalized features.
- 5) **My Account Page:** Allows registered users to manage their profiles, update personal information, and view their course registration details.
- 6) **Course Search and Browsing:** Allows users to search and filter available courses based on relevant criteria.
- 7) **Announcements Page:** Provides important updates, notices, and information related to short courses.
- 8) **Admin Management:** Allows authorized university staff to manage courses, registrations, users, announcements, and other website content.
- 9) **Contact Us Page:** Provides contact information and a communication form for learners to submit inquiries and feedback.
- 10) **About Us Page:** Provides information about the Open University of Sri Lanka, Kandy Regional Centre, and its short-course services

7.2.3. Scope Exclusions

The current phase of the **Open University Short Course Website (KRC)** will not include the following elements:

- 1) **Standalone Mobile Applications:** Development of separate Android or iOS applications is not included. The website will instead be responsive and accessible on mobile devices.
- 2) **Offline Accessibility:** Users will require an internet connection to access the website and its services. Offline access is not included.
- 3) **Multilingual Support:** The initial version will be developed in a single language, most likely English. Multilingual support may be considered in future phases.

- 4) **Full Online Learning/LMS Features:** The current system will focus on short-course information and registration rather than providing complete online classes, examinations, and learning management features.
- 5) **Advanced AI Services:** AI-based learning assistance and advanced chatbot features are not included in the initial development phase

7.2.4. Impact on Other Systems

The **Open University Short Course Website (KRC)** will have several impacts on existing systems and processes, including:

- 1) **Integration with University Systems:** The website may interact with existing university databases and systems to manage course information, student details, and registration records efficiently.
- 2) **Data Security and Privacy:** The system will implement secure authentication, data protection, and access controls to protect learners' personal and registration information.
- 3) **Administrative Processes:** The website will reduce manual work by digitizing course promotion, application, registration, and information management processes, improving the efficiency of university staff.
- 4) **Student Engagement:** By providing centralized and easily accessible course information, the website will improve communication and engagement between learners and the university.
- 5) **Communication and Notifications:** The system can support announcements and notifications related to courses, registration deadlines, schedules, and other important university information.
- 6) **Educational Accessibility:** Learners will be able to access course information and registration services from any location with an internet connection, making short courses more accessible.
- 7) **Future System Integration:** The website can be expanded in future to integrate with online payment systems, learning management systems (LMS), email/SMS notification services, and other university platforms

7.3. Functional Requirements

The functional requirements for the **Open University Short Course Website (KRC)** are detailed below, outlining the essential features and functionalities that the system must provide to meet the

needs of learners and university staff in managing and accessing short-course information and registration services.

7.3.1. User Access and Engagement

- 1) The website shall provide a **welcoming home page** displaying available short courses, important announcements, and upcoming courses.
- 2) Users shall be able to **log in** to access personalized features, manage their profiles, and view their course registrations.
- 3) New users shall be able to **create an account** to access registration and other available features, while guest users shall have limited access to browse course information.
- 4) The system shall provide a **Forgot Password** feature that allows users to securely reset their password by verifying their account details.
- 5) The system shall allow registered users to **update their profile information**, including personal and contact details.
- 6) The system shall provide **secure logout functionality** to protect user accounts.

7.3.2. Information and Context Management

- 1) The **Course Information Page** shall be regularly updated with the latest information about available short courses, including course descriptions, schedules, fees, and eligibility requirements.
- 2) A **Course Details Page** shall provide comprehensive information about each course to help learners make informed decisions before applying.
- 3) The **Announcements Page** shall display important notices, registration deadlines, course updates, and other relevant information for learners.
- 4) The **Upcoming Courses Page** shall list upcoming short courses, workshops, training programmes, and other educational activities offered by the Kandy Regional Centre.

7.3.3. Education Resources and Tools

- 1) The **Course Registration Page** shall provide a step-by-step online registration process that allows learners to apply for selected short courses easily and efficiently.
- 2) The **Course Selection and Application Page** shall provide practical guidance and required information to help learners select suitable courses based on their interests, eligibility, and requirements.
- 3) The **Fee and Payment Information** shall provide details about course fees and payment requirements to support learners in planning their course enrollment.

- 4) The system shall allow learners to **view their registration status and application details** through their user account.

7.3.4. User Interaction and Support

- 1) The **About Us** page shall provide information about the **Open University of Sri Lanka – Kandy Regional Centre (KRC)**, its short-course services, objectives, and relevant information to build trust and transparency.
- 2) The **Contact Us** page shall provide contact information and a communication form to allow learners to submit inquiries, requests, and feedback.
- 3) The **My Account** page shall allow registered users to manage their personal profiles and view their course registration information.
- 4) Users shall be able to **update their email address and password** through the My Account section.
- 5) Users shall be able to **upload and update their profile picture**.
- 6) The system shall ensure that only authenticated users can access and modify their personal account information.

7.3.5. Contribution to the University

The **Open University Short Course Website (KRC)** aims to centralize and provide easy access to short-course information and registration services through a single online platform. By reducing manual processes, improving communication, and making course information more accessible, the system will enhance the efficiency of the **Open University of Sri Lanka – Kandy Regional Centre**. It will also improve the learner experience, increase awareness of available short courses, and support the university in expanding and effectively managing its short-course programmes.

7.3.6. Data Archival And Retention

The **Data Archival and Retention** functionality of the **Open University Short Course Website (KRC)** ensures that important course, registration, user, and administrative data is systematically stored and maintained. This includes:

- 1) **Archival Process:** The system shall archive completed course, registration, and relevant administrative records at regular intervals to maintain historical information.
- 2) **Retention Policies:** The system shall retain data according to the university's data retention policies and applicable legal or regulatory requirements.
- 3) **Access to Archived Data:** Authorized university administrators shall be able to retrieve and review archived course and registration records when required.

- 4) **Data Security:** Archived data shall be securely stored and protected using appropriate security measures to prevent unauthorized access, modification, or loss

7.4. User Profiles, Roles and Privileges

Requirement No.	Requirement Name	Description	Error Handling	Priority
01	User Registration and Authentication	•Registration: Users can register by providing the required personal and account details. • Login: Registered users can log in using their email address and password. • Password Reset: Users can reset their password if they forget it by following the verification process.	• If a user attempts to log in without an account, the system shall display “Register Before Login” and direct the user to the registration page. • If an invalid email or password is entered, an appropriate error message shall be displayed.	High
02	View Home Page	Users can access the home page to view available short courses, latest announcements, upcoming courses, and important information.	If required content cannot be loaded, the system shall display an appropriate error message.	High
03	User Profile Management	• Users can manage their profiles and update personal information. • Users can change their email address and password.	• Users must enter the current password before changing their password. • If invalid information is entered, an appropriate error	Medium

Requirement No.	Requirement Name	Description	Error Handling	Priority
		• Users can upload or update their profile picture. • Users can view their course registration details.	message shall be displayed.	
04	View Short Courses	• Users can browse available short courses. • Users can search and filter courses based on relevant information • Users can view detailed course information such as description, duration, fees, schedule, and eligibility.	If no course matches the search criteria, the system shall display “No Courses Found.”	High
05	View Course Details	Users can view complete information about a selected short course, including course description, objectives, duration, schedule, fees, eligibility requirements, and registration information.	If course information is unavailable, the system shall display an appropriate message.	High
06	Course Registration	• Registered users can apply for available short courses through the website. • Users can submit the required registration information.	• Only logged-in users can register for courses. • If required fields are incomplete, the system shall display an error message. • Users shall not be allowed to register	High

Requirement No.	Requirement Name	Description	Error Handling	Priority
		Users can view their registration status after submitting an application.	for unavailable or closed courses.	
07	View Announcements and Updates	Users can view important university announcements, registration deadlines, course updates, and other information related to short courses.	If announcements cannot be loaded, the system shall display an appropriate error message.	Medium
08	View Upcoming Courses and Events	Users can view upcoming short courses, workshops, training programmes, and other educational activities organized by the Kandy Regional Centre.	If no upcoming courses or events are available, the system shall display “ No Upcoming Events Available. ”	Medium
09	Manage Course Registrations	Authorized university staff can view, manage, approve, reject, and update learner registration records.	If an invalid registration record is selected or updated, the system shall display an appropriate error message.	High
10	Admin Course Management	Administrators can add, update, delete, and manage short-course information, including course descriptions, fees, schedules, and eligibility requirements.	The system shall validate required information before allowing course details to be saved or updated.	High

Requirement No.	Requirement Name	Description	Error Handling	Priority
11	About Us	Users can view information about the Open University of Sri Lanka – Kandy Regional Centre , its short-course services, objectives, and relevant information.	If the information cannot be retrieved, the system shall display an appropriate error message.	Low
12	Contact Us	Users can view contact information and submit inquiries, questions, and feedback to the university.	- Required fields must be completed before submission. - If invalid information is entered, the system shall display an appropriate error message.	Low
13	User Registration Status	Registered users can view the status of their submitted short-course applications and registration details through their accounts.	If no registration record exists, the system shall display “No Registration Found.”	Medium
14	Notifications and Updates	The system can provide users with important notifications regarding registration confirmations, course updates, deadlines, and other relevant information.	If a notification cannot be delivered, the system shall record the issue and display an appropriate message where applicable.	Medium

Table 2: User Profiles ,Roles and Privileges

7.4.1. Use Case Diagram

This Use Case Diagram illustrates the main interactions between the Open University Short Course Website and its three **actors: Guest User, Student/Learner, and Administrator.**

Guest users can browse and search publicly available course information. **Students** can register, log in, register for courses, view course details, announcements, events, learning resources, timetables, and manage their profiles. **Administrators** can manage users, registrations, courses, programmes, announcements, events, resources, system settings, and reports.

The diagram also shows relationships between functions such as registration, login, course registration, payment processing, and registration management. Overall, this diagram demonstrates how the system supports efficient online course information, registration, and administration.

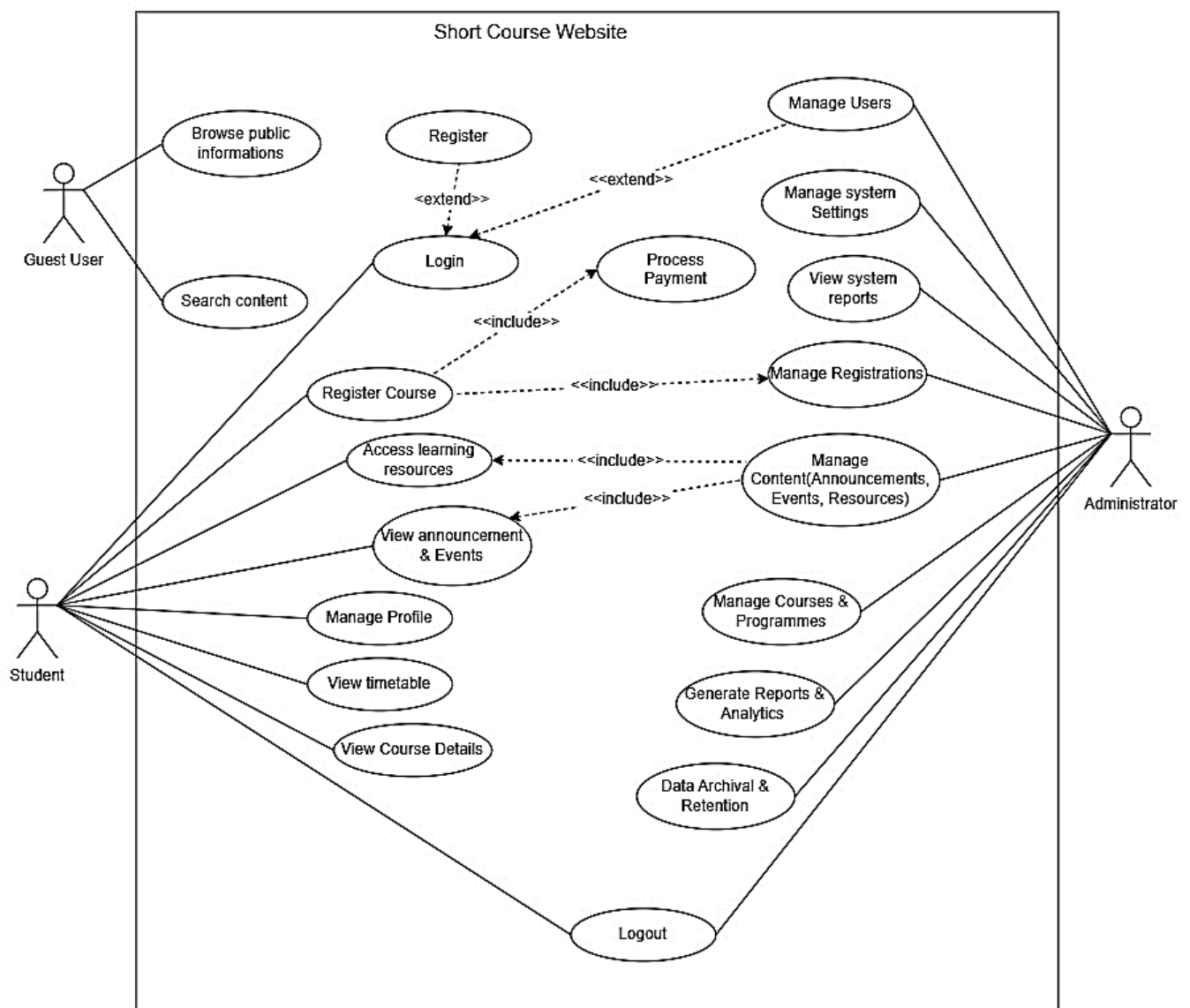


Figure 4: Use Case Diagram

7.4.2. Main Actors of the System

Sr. No.	Actor	Role
1.	Student / Learner	Registers for an account, browses courses, views course details, applies for courses, and manages their profile.
2.	Guest User	Browses publicly available course information, announcements, and university information without logging in.
3.	Administrator / University Staff	Manages courses, student registrations, announcements, users, and other website content.

7.4.3. 7.4.4. Reporting Requirements

The **Reporting Requirements** for the **Open University Short Course Website (KRC)** involve generating reports that provide useful information about course registrations, users, and website activities. Such as:

- 1) **Course Registration Reports:** Reports on the number of registrations for each short course, helping administrators identify popular courses and enrollment trends.
- 2) **User Reports:** Information about registered users, active users, and user registration details to support effective user management.
- 3) **Course Performance:** Reports showing the number of applications and registrations for each course to evaluate course demand and performance.
- 4) **Registration Status Reports:** Reports on approved, pending, and rejected course applications, helping university staff monitor the registration process.
- 5) **Administrative Reports:** Reports on courses, announcements, registrations, and other system activities to support decision-making and efficient management.
- 6) **Data Visualization:** Graphical representation of registration and course data to help administrators easily understand trends and make informed decisions

7.5. Non Functional Requirements

The non-functional requirements define the quality, performance, security, usability, and reliability standards that the **Open University Short Course Website (KRC)** should meet.

Requirement No.	Requirement Name	Description	Priority
NFR 01	Performance	The website shall load pages and process user requests within a reasonable time under normal conditions.	High
NFR 02	Security	The system shall protect user accounts and personal information using secure authentication, password protection, and access control.	High
NFR 03	Usability	The website shall provide a simple, clear, and user-friendly interface that allows learners to easily find courses and complete registrations.	High
NFR 04	Availability	The website shall be available whenever users need to access course information and registration services, except during scheduled maintenance.	High
NFR 05	Reliability	The system shall operate consistently and accurately without unexpected failures or loss of registration data.	High
NFR 06	Scalability	The system shall be capable of supporting an increasing number of users, courses, and registrations as the university expands its short-course services.	Medium
NFR 07	Compatibility	The website shall work correctly on commonly used web browsers such as Chrome, Edge, Firefox, and Safari.	Medium
NFR 08	Responsiveness	The website shall provide a responsive interface that works effectively on desktops, tablets, and mobile devices.	High

Requirement No.	Requirement Name	Description	Priority
NFR 09	Maintainability	The system shall be designed so that administrators and developers can easily update course information, announcements, and other content.	Medium
NFR 10	Data Integrity	The system shall ensure that user, course, and registration data remains accurate, consistent, and protected from unauthorized modification.	High
NFR 11	Backup and Recovery	The system shall regularly back up important data and provide mechanisms to restore data in case of system failure or data loss.	High
NFR 12	Accessibility	The website shall use clear text, readable layouts, and appropriate navigation to make the system accessible to a wide range of learners.	Medium
NFR 13	Privacy	The system shall protect users' personal information and ensure that it is accessed only by authorized users.	High
NFR 14	Extensibility	The system shall allow future integration with services such as online payment, email/SMS notifications, and learning management systems.	Medium

Table 3: Non Functional Requirements

The non-functional requirements of the **Open University Short Course Website (KRC)** define the overall quality and performance standards that the system should maintain beyond its basic functions. The website should be **secure, reliable, efficient, user-friendly, and accessible** to provide a positive experience for both learners and university staff. Good system performance is important because users should be able to access course information, navigate between pages, and complete registrations without unnecessary delays.

Security and privacy are also important because the system will handle personal information such as learner names, email addresses, contact details, login credentials, and registration

information. Appropriate authentication, authorization, password protection, and access controls should be implemented to prevent unauthorized access. The system should also maintain **data integrity** so that course and registration information remains accurate and consistent.

The website should provide **high availability and reliability**, allowing learners to access course information and registration services whenever required, except during planned maintenance. Regular **backup and recovery mechanisms** should be implemented to reduce the risk of losing important university and learner data. The system should be **responsive and compatible** with commonly used web browsers and devices, including desktop computers, tablets, and mobile phones. A simple and user-friendly interface should make it easy for learners to search for courses, view course details, register for courses, and manage their accounts.

The system should also be **maintainable and scalable**, allowing administrators and developers to update course information and other content easily while supporting an increasing number of users and courses. Finally, the system should be **extensible** so that future features such as **online payment facilities, email/SMS notifications, analytics, and Learning Management System (LMS) integration** can be added as the university's requirements develop.

7.6. User Interfaces

The **Open University Short Course Website (KRC)** shall provide a simple, modern, and user-friendly interface that allows learners and university staff to access the system easily. The interface will be designed to provide clear navigation and organized information so that users can quickly find available short courses, course details, schedules, fees, eligibility requirements, and registration information.

The **Home Page** will provide an overview of available courses, important announcements, upcoming courses, and quick links to frequently used sections. The **Course Listing and Course Details pages** will allow users to browse and view detailed information about each short course. The **Registration interface** will provide a clear step-by-step form for learners to submit their course applications. Registered users will have access to a **Login, Registration, and My Account interface**, where they can manage their personal information, update their password, upload a profile picture, and view their registration status. The **Administrator interface** will provide authorized university staff with dashboards for managing courses, users, registrations, and announcements. The website will use a **responsive design** so that all interfaces can be accessed effectively on desktop computers, tablets, and mobile devices. Consistent layouts, readable fonts, clear buttons, appropriate forms, and meaningful error messages will be used throughout the system to provide a smooth and accessible user experience.

7.6.1. WireFrame

7.6.1.1. Guest homepage

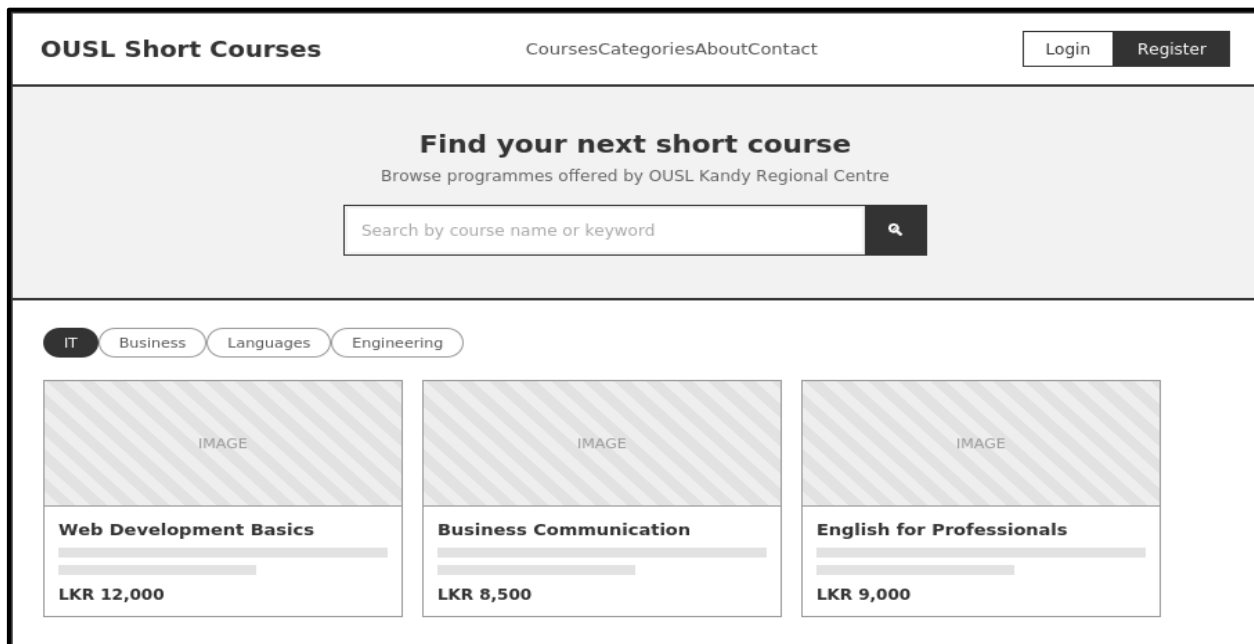


Figure 5: Homepage wireframe (Guest User) -course browsing, search, and login/register

7.6.1.2. Student dashboard

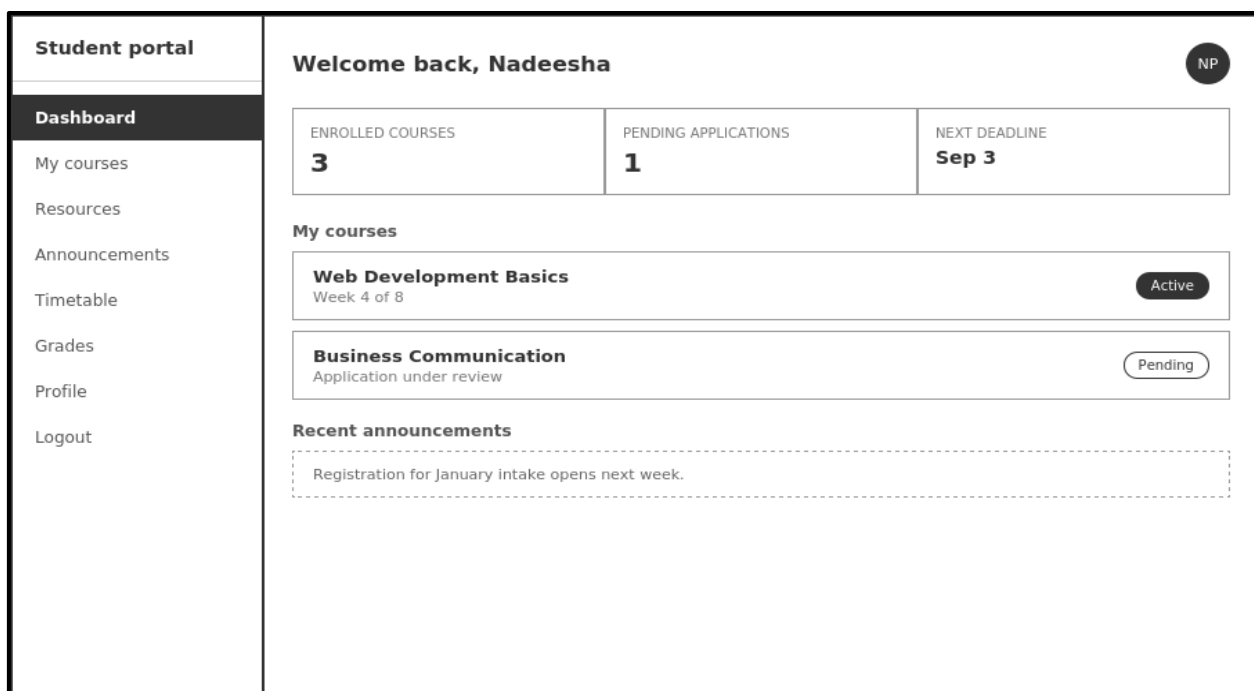


Figure 6: Administration dashboard wireframe -user management, courses, enrollment tracking, and reports

7.6.2.1. Administration dashboard

Administration	Overview
Overview	
Manage users	
Courses and programmes	
Enrollment tracking	
Content and announcements	
Reports and analytics	
Data archival	
System settings	

ACTIVE COURSES	TOTAL LEARNERS	PENDING APPLICATIONS	NEW THIS WEEK
24	612	37	58

Recent applications [View all](#)

Applicant	Course	Status
R. Fernando	Web Development	Pending
S. Perera	Business Comm.	Approved

Quick actions

+ Add course	Post announcement	Generate report
--------------	-------------------	-----------------

Figure 7: Administration dashboard wireframe -user management, courses, enrollment tracking, and reports

The following additional screens cover the remaining navigation items shown in the sidebars above

7.6.2.2. Login and register

Log in to your account

Email

Password

Log in

Don't have an account? [Register](#)

Figure 8: Login / register screen (Guest User)

7.6.2.3. Student - learning resources

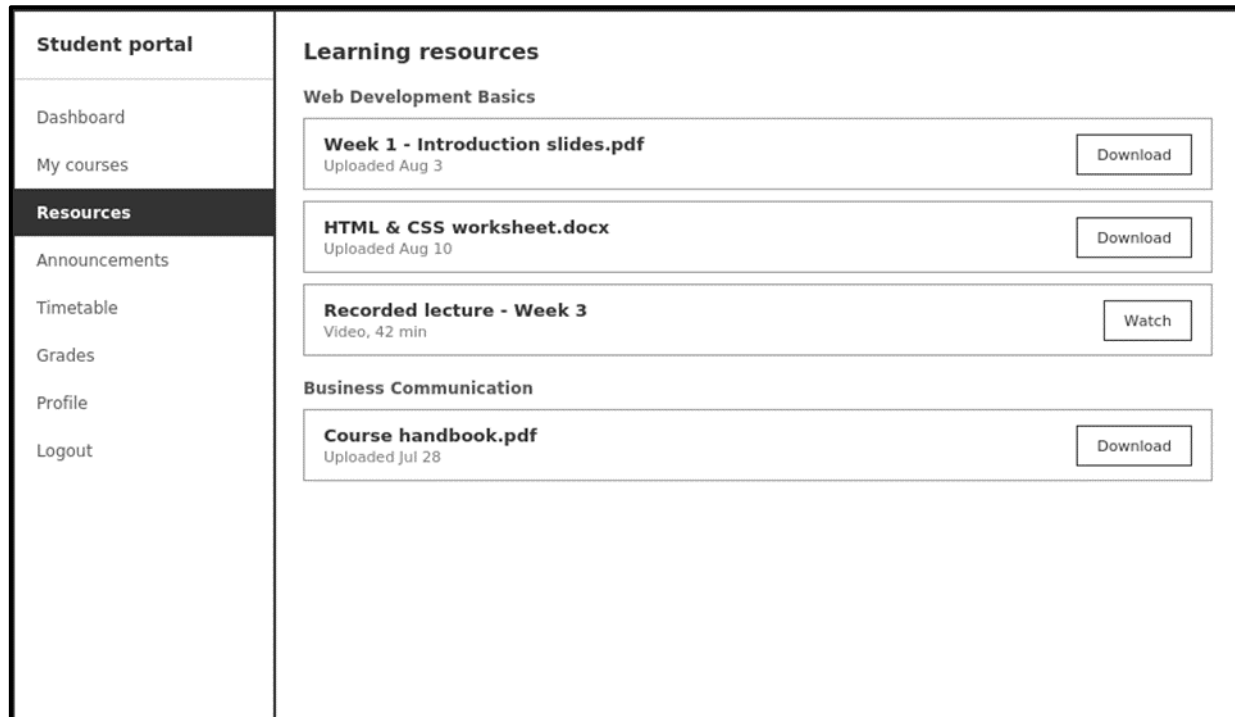


Figure 9: Resources screen — downloadable materials and recordings per enrolled course

7.6.2.4. Student – timetable

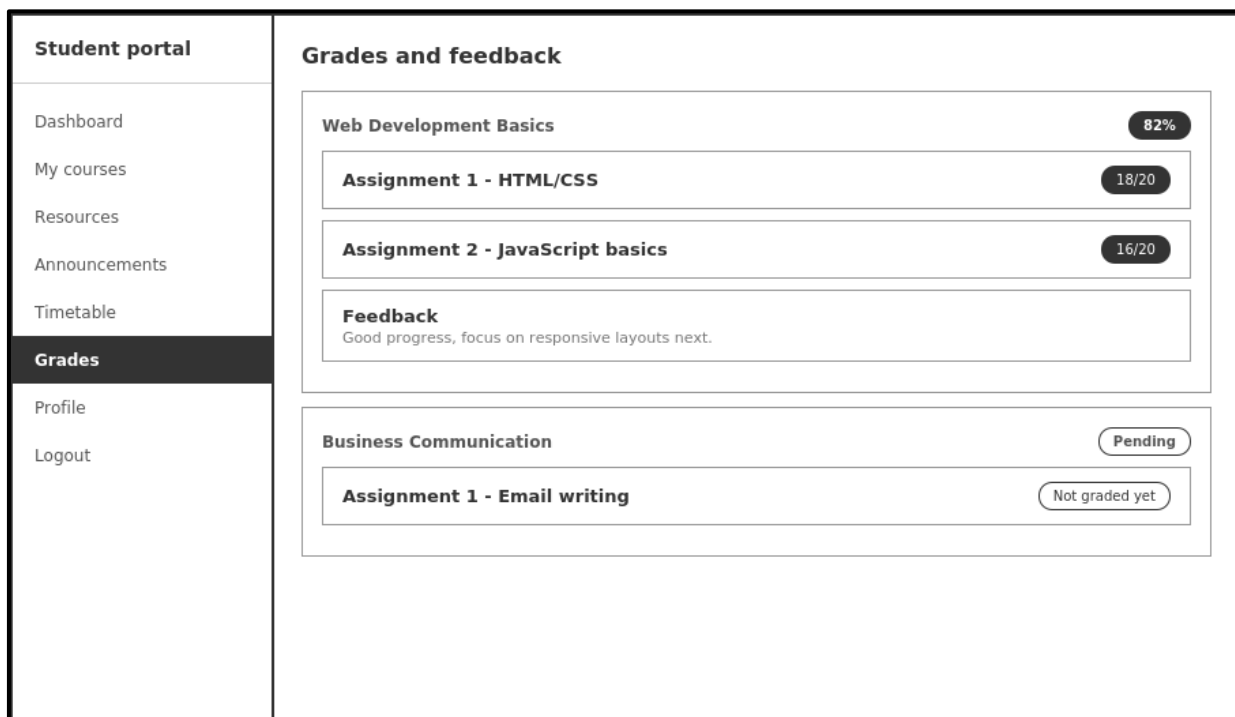


Figure 10: Profile screen - editable personal details and password change

7.6.2.5. Admin - manage users

Administration	Manage users
Overview	+ Add user Export list
Manage users	
Courses and programmes	
Enrollment tracking	
Content and announcements	
Reports and analytics	
Data archival	
System settings	

Name	Role	Status	Actions
Nadeesha Perera	Student	Active	Edit / Suspend
R. Fernando	Student	Active	Edit / Suspend
K. Silva	Administrator	Active	Edit / Suspend
S. Perera	Student	Suspended	Edit / Reactivate

Figure 11: Manage users screen -account list with role and status controls

7.6.2.6. Admin - courses and programmes

Administration	Courses and programmes
Overview	+ Add course
Manage users	
Courses and programmes	
Enrollment tracking	
Content and announcements	
Reports and analytics	
Data archival	
System settings	



Web Development Basics

Edit

8 weeks - Online - 42 enrolled



Business Communication

Edit

6 weeks - Blended - 28 enrolled



English for Professionals

Edit

10 weeks - Physical - 19 enrolled

Figure 12: Courses and programmes screen — course list with enrollment counts

7.6.2.7. Admin - content and announcements

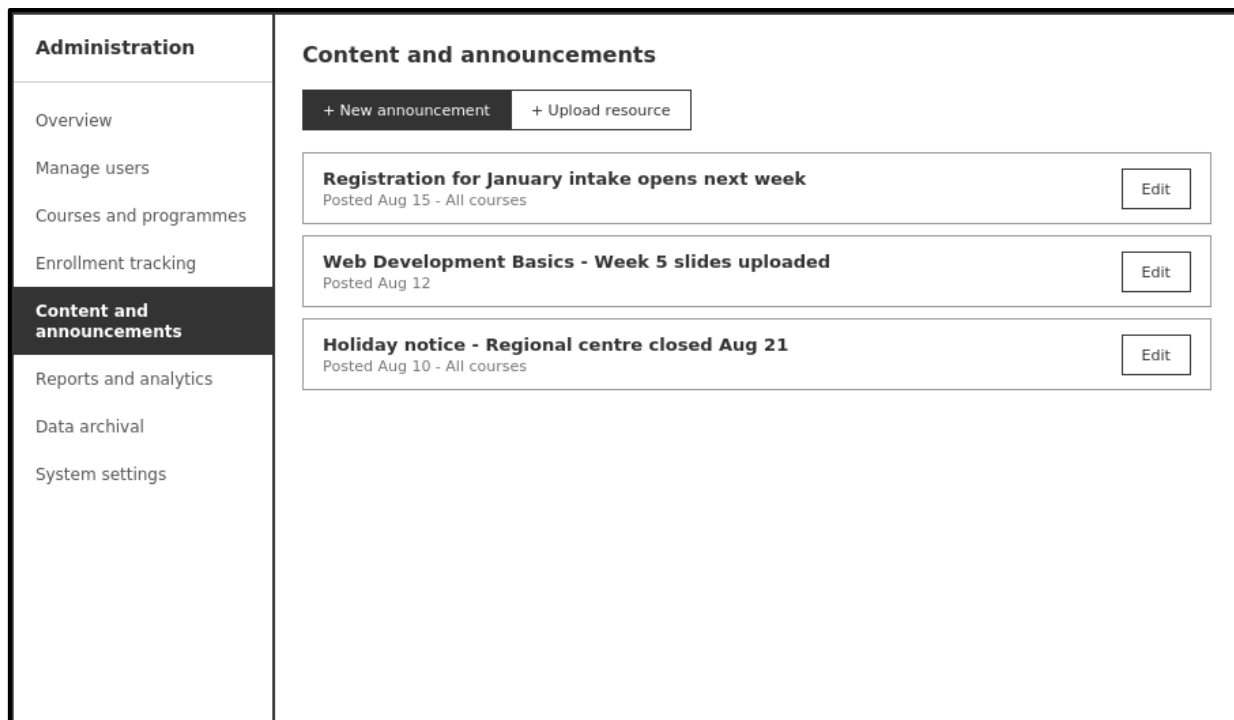


Figure 13: Content and announcements screen — published items with edit controls

7.6.2.8. Admin - reports and analytics

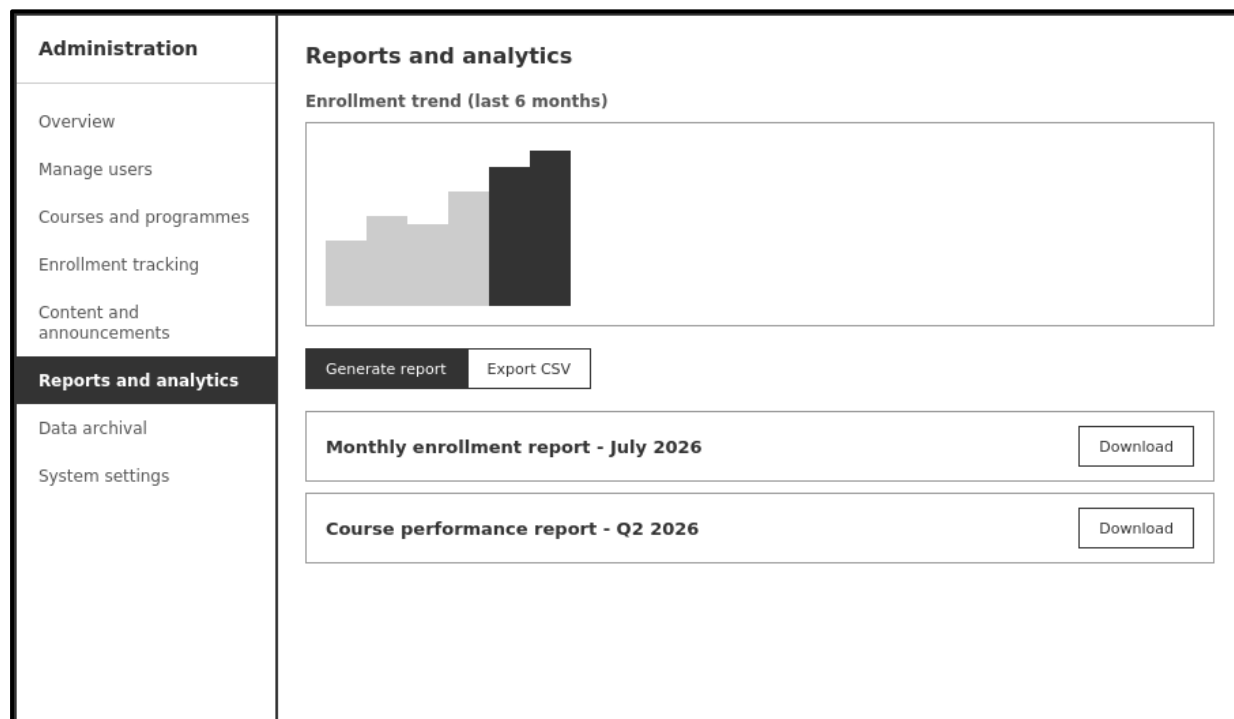


Figure 14: Reports and analytics screen -enrollment trend and downloadable reports

7.6.2.9. Admin - system settings

Administration	System settings
Overview	
Manage users	
Courses and programmes	
Enrollment tracking	
Content and announcements	
Reports and analytics	
Data archival	
System settings	

Site name

Support email

Registration period defaults

Data retention period

Save settings

Figure 15: System settings screen - site-wide configuration fields

7.6.2.10. Prototype Links

<https://www.figma.com/design/CCC0ffR79NZeKDy89AvXfL/Design-Group-Project?node-id=1-2&t=aZ5XKYSox9Jmky7o-1>

7.7. Hardware Interfaces

The **Open University Short Course Website (KRC)** will require appropriate hardware resources to support the web application, database, and users accessing the system. The main hardware interfaces are as follows:

7.7.1. Server Requirements

Description: The server hosts the website backend, application services, and database.

Specifications:

1. **Processor:** Minimum Quad-Core 2.5 GHz
2. **RAM:** Minimum 16 GB
3. **Storage:** Minimum 1 TB SSD
4. **Network:** Gigabit Ethernet
5. **Operating System:** Windows Server or Linux-based server

7.7.2. Client Devices

Description: Devices used by learners, guests, and university staff to access the website.

Specifications:

1. **Desktops/Laptops:** Minimum Dual-Core 2 GHz processor
2. **RAM:** Minimum 4 GB
3. **Storage:** Minimum 128 GB
4. **Display:** Minimum resolution of 1366 × 768
5. **Network:** Ethernet or Wi-Fi connectivity

7.7.3. Web Application Server

Description: Hosts the website application and processes requests from users.

Specifications:

1. **Type:** Web/Application Server
2. **Processor:** Minimum Quad-Core 2.5 GHz
3. **RAM:** Minimum 8 GB
4. **Storage:** Minimum 500 GB SSD for application files and logs
5. **Network:** High-speed internet connection

7.7.4. Database Server

Description: Stores and manages learner, course, registration, and administrative data.

Specifications:

1. **Type:** Database Server
2. **Processor:** Minimum Quad-Core 2.5 GHz
3. **RAM:** Minimum 8 GB
4. **Storage:** Minimum 1 TB SSD
5. **Database:** MongoDB
6. **Backup:** Automated regular backups with secure storage

7.7.5. Network Interface

Description: : Provides communication between users, the web server, and database services.

Specifications:

1. **Connection:** Ethernet or Wi-Fi
2. **Internet:** Reliable broadband internet connection
3. **Network Speed:** Minimum 100 Mbps recommended for the server environment
4. **Security:** Firewall and secure network configuration

7.8. Software Interfaces

The **Open University Short Course Website (KRC)** will use the following software interfaces to support development, deployment, and access to the system.

7.8.1. Operating System Requirements

The website shall be accessible through commonly used operating systems, including:

1. **Windows**
2. **Android**
3. **macOS**
4. **iOS**

7.8.2. Development Tools and Software

The system will be developed and maintained using the following software:

1. **Visual Studio Code (VS Code)** – Code editor for frontend and backend development.

2. **Node.js** – Runtime environment for the backend.
3. **React.js** – Frontend development framework.
4. **Express.js** – Backend web application framework.
5. **MongoDB** – Database management system.
6. **Git/GitHub** – Version control and source code management.
7. **Figma** – User interface and prototype design.
8. **Draw.io** - Usecase diagram, context diagram & flow chart
9. Web Browsers

Users shall be able to access the website through commonly used web browsers, including:

1. Google Chrome
2. Microsoft Edge
3. Mozilla Firefox
4. Safari

7.9. Communication Interfaces

The **Open University Short Course Website (KRC)** will use several communication interfaces to support interaction between learners, university staff, and the system.

1. **Email:** Email services may be used for registration confirmations, password recovery, course updates, and important notifications.
2. **Internet/Network:** The website will communicate with users through an internet connection using standard **HTTP/HTTPS** protocols. HTTPS will be used to provide secure communication between the user's device and the website.
3. **Contact Forms:** The Contact Us interface will allow learners and visitors to submit inquiries, feedback, and support requests directly to the university.
4. **Future Notification Services:** The system can be extended to support SMS or other notification services for important course and registration updates.

7.10. Security and Authentication Requirements

The Open University Short Course Website (KRC) shall implement appropriate security and authentication mechanisms to protect learner, administrator, course, and registration information. Since the system handles personal user information and registration records, security is an important requirement.

7.10.1. User Authentication

The system shall require users to authenticate using a registered email address and password before accessing protected features such as course registration and account management.

7.10.2. User Registration

New users shall be required to provide valid personal and account information when creating an account. The system shall validate the provided information before creating the account.

7.10.3. Password Security

User passwords shall be stored securely using password hashing rather than storing passwords as plain text. The system shall also enforce suitable password requirements to improve account security.

7.10.4. Password Reset

The system shall provide a Forgot Password function. Users who forget their passwords shall be able to reset them through a secure verification process, such as an email-based reset link or verification mechanism.

7.10.5. Role-Based Access Control

The system shall provide different access permissions based on user roles:

1. Guest: Can view publicly available course and university information.
2. Student/Learner: Can register for courses, manage their profile, and view registration details.
3. Administrator: Can manage users, courses, registrations, announcements, and other system content.

7.10.6. Authorization

The system shall verify user permissions before allowing access to restricted pages or functions. For example, only authorized administrators shall be able to modify course information or manage student registrations.

7.10.7. Secure Communication

The website shall use HTTPS/SSL/TLS to encrypt communication between users' devices and the server. This helps protect login credentials and personal information from unauthorized interception.

7.10.8. **Session Management**

The system shall securely manage user sessions and automatically terminate or expire inactive sessions after a defined period. Users shall also be provided with a secure Logout option.

7.10.9. **Input Validation**

The system shall validate and sanitize user inputs to prevent security threats such as SQL/NoSQL injection, cross-site scripting (XSS), and malicious form submissions.

7.10.10. **Account Protection**

The system should limit repeated unsuccessful login attempts to reduce the risk of brute-force attacks. Appropriate error messages should be provided without revealing sensitive account information.

7.10.11. **Data Privacy**

Personal information such as names, email addresses, contact details, and registration records shall only be accessible to authorized users. The system shall not unnecessarily expose or share user information

7.10.12. **Administrative Security**

Administrator accounts shall have stronger access controls because administrators can modify important system information. Administrative functions shall only be accessible to authenticated and authorized university staff.

7.10.13. **Backup and Recovery**

Important user, course, and registration data shall be backed up regularly. Backup data shall also be protected from unauthorized access and used to recover the system in case of data loss or failure.

7.10.14. **Security Monitoring**

The system should maintain appropriate logs of important activities, such as administrator logins, registration updates, and changes to course information. These logs can help identify and investigate suspicious activities.

8. Appendix

8.1. Letter Requesting Approval

Date: 7/14/2026

To:

The Assistant Director
The Open University of Sri Lanka
Kandy Regional Centre
Kandy

Subject: Request for Approval to Develop a Website for Short Courses as the EER4189 Software Designing Group Project

Dear Sir/Madam,

We are a group of undergraduate students from the Open University of Sri Lanka, Kandy Regional Centre, currently following the course EER4189 – Software Designing. As part of the course requirements, we are required to complete a group software development project.

Our group has selected the development of a Website for the Short Courses offered by the Open University of Sri Lanka, Kandy Regional Centre as our project. The proposed website aims to provide an organized and user-friendly platform that will enable prospective students to view available short courses, obtain course information, access application details, and improve the overall accessibility of short course information.

Since this project is based on the services provided by the Kandy Regional Centre, we kindly request your approval to undertake this project. We assure you that this project is intended solely for academic purposes, and any information collected or used will be handled responsibly and only with the necessary permission. We will also ensure that the University's policies and guidelines are respected throughout the development process.

We would be grateful if you could kindly grant us permission to proceed with this project. Your approval will enable us to carry out the necessary requirement gathering, system analysis, design, and implementation activities successfully.

Thank you for your time and consideration. We look forward to your favorable response.

Yours faithfully,

Group Members

Group Leader

I.I.K. Wimalasekara

Registration No: S92068547

1. K. Pushpakanthan
Registration No: 121421424
2. M.M.A.G.S.A. Warnathilaka
Registration No: 422514144
3. R.K.T.D. Gunasekara
Registration No: 62251610

Figure 16: Letter Requesting Approval